

Assignment 3

Model Training & Security Evaluation

From baseline to adversarial attack

Due: **Friday 12 June 2026, 11:59 PM**

Worth: **25%**

Assignment Overview

Building on your A1 and A2 baseline model, Assignment#3 takes you through four connected steps. From properly training a Model to stress-testing it with adversarial attacks. Use your dataset from Assignment#2

Report Structure and what to include in Your Assignment#3.PDF Report

Step 1:- Baseline Results (from A1 / A2) -----→[15 pts]

Include screenshots showing your model before any changes. These are your **BEFORE** numbers and label them clearly as '**Baseline**' in your report.

What to show:

1. Screenshot of the classification report (accuracy, precision, recall, F1)
2. Screenshot of the confusion matrix
3. Weighted average F1 score noted clearly
4. Training time (add %%time to your training cell)

Step 2:- Fine-Tuning & Improvements -----→ [30 pts]

Show the changes you made to improve the model and compare results directly against your baseline.

What to show:

1. Feature importance bar chart (top features)
2. Number of features kept vs dropped (and why they are dropped?)
3. Best hyperparameters found by GridSearchCV (if used)
4. Before vs After comparison table: F1 (weighted), accuracy, features used
5. Screenshot of improved classification report

Metric	Baseline (your Model Name)	Fine-tuned (your Model Name)	Change
F1	your score	new score	up (improved) or down
Accuracy	your score	new score	up (improved) or down
Features used	78	30	down = 48 features

Step 3:- Prediction Pipeline Demo -----→ [30 pts]

Show your model making live predictions on real rows from your test set. (I will **Demo** during our class)

What to show:

1. 3 prediction examples: 1 BENIGN + 2 different attack types
2. For each: real label, predicted label, and confidence score
3. Any surprising or incorrect predictions noted and briefly explained

```
# Example: pull a real row from your test set
sample_idx = 390 # change this to try different rows
sample_row = pd.DataFrame(X_test_imp[sample_idx:sample_idx+1], columns=feat_names)
sample_dict = sample_row.iloc[0].to_dict()

print(f'Real label : {y_test.iloc[sample_idx]}')
predict_traffic(sample_dict)
```

*** Real label : Web Attack ⚡ Brute Force
Prediction : Web Attack ⚡ Brute Force
Confidence : 69.0%

Top 3 probabilities:

Web Attack ⚡ Brute Force	0.69
Web Attack ⚡ XSS	0.31
BENIGN	0.00

('Web Attack ⚡ Brute Force',
Web Attack ⚡ Brute Force 0.69
Web Attack ⚡ XSS 0.31
BENIGN 0.00
Bot 0.00
DoS GoldenEye 0.00
DDoS 0.00
DoS Hulk 0.00
DoS Slowhttptest 0.00
FTP-Patator 0.00
DoS slowloris 0.00
SSH-Patator 0.00
PortScan 0.00
dtype: float64)

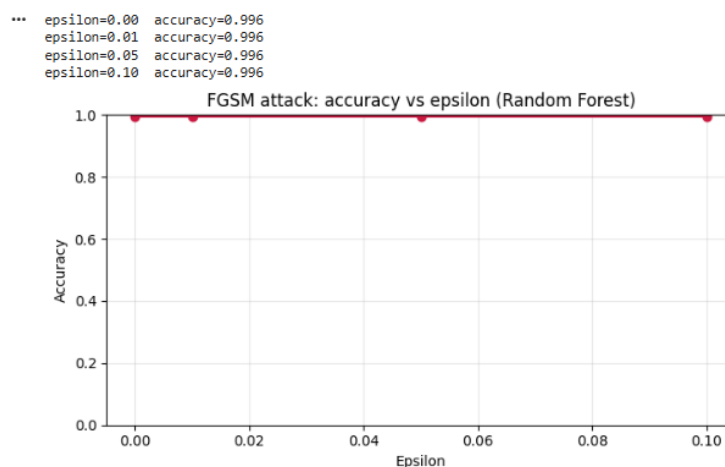
Step 4:- FGSM Adversarial Attack & Defence -----→ [15 pts]

FGSM nudges each feature to increase model error. Small changes to packet sizes, flow duration, and byte counts can cause the model to misclassify attacks as BENIGN.

Note: You are free to use any adversarial attack “FGSM” is just the example I used in the notebook.
Check the quick reference table in your Week 5.1 lecture to find the attack that best fits your model (e.i. ZOO for Random Forest, TextFooler for NLP and C&W for CNN).

What to show:

1. Test 4 epsilon values: 0.00 (no attack), 0.01, 0.05, 0.10
2. Accuracy-vs-epsilon line chart (accuracy on Y, epsilon on X)
3. One defence implemented (Input Preprocessing, Feature Squeezing, or Adversarial Training)
4. Before vs After defence metrics table (accuracy & F1 at epsilon=0.05)
5. Written explanation: did the defence work? What trade-offs did it introduce?



Your written report must cover all 5 sections below. Be specific and use actual numbers, not vague claims.

#	Section	What to include
1	Model & Dataset	Dataset name, security problem, baseline RF metrics, confusion matrix screenshot
2	Fine-Tuning Results	Key features kept, features dropped, best hyperparameters, before vs. after F1
3	Prediction Demo	3 examples (BENIGN + 2 attacks): real label, predicted label, confidence. Any surprises?
4	FGSM Attack Results	Accuracy vs epsilon table or chart. At which epsilon did it break? Real-world attacker risk?
5	Defence & Reflection	Which defence? Before/after metrics. Trade-offs. Which would you deploy in production and why?

Marking Rubric: 100 points = 25%

Step	Description	Points
1 Baseline	Classification report + confusion matrix present. Baseline accuracy & F1 recorded.	15
2 Fine-Tuning	Feature importance chart shown. GridSearchCV run. Before/after F1 compared.	30
3 Prediction	Predict traffic works using 3 examples shown with confidence scores.	30
4 FGSM & Defence	Attack at 4 epsilon values. Accuracy chart. Defence with before/after metrics.	15
Code.ipynb file	Full code for your project (Missing the file -100%)	10